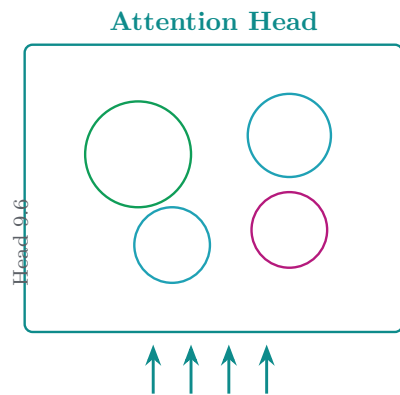


BEYOND COMPONENTS

Recovering Scalable Algorithmic Circuits via Low-Rank Discovery

‘Core Engine’

1. The problem — the ‘atomic’ fallacy



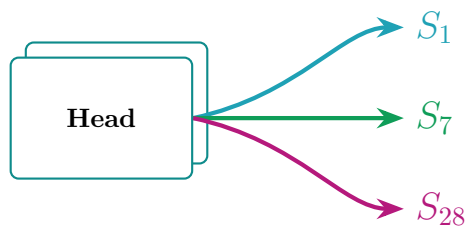
Polysemanticity — *one component doing many jobs at once.*

Different-coloured gears = unrelated features all mixed inside the same head (gender, number, punctuation, sentiment ...).

Existing interpretability methods treat components (neurons, heads) as **indivisible ‘atoms,’** which masks the true computational graph.

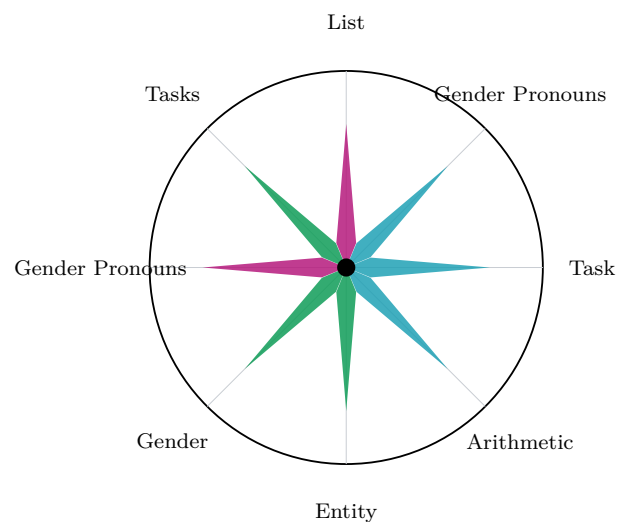
If one head handles gender, number, punctuation *and* sentiment at once, “this head is responsible for X ” is not a meaningful statement. We need to go one level deeper.

2. The solution — SVD deconstruction



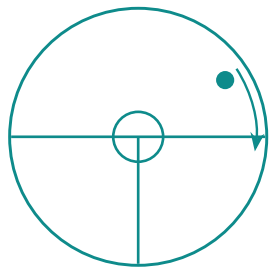
SVD

Mechanism. SVD of the head’s output-value matrix $W_{OV} = U\Sigma V^\top$ gives an ordered basis. **Each row of $V^\top$ is one direction the head can write.** Top rows often correspond cleanly to a single concept.

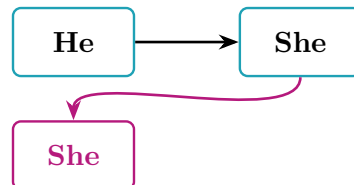


Singular directions point to specific concepts — one direction, one named task.

3. The proof — surgical control

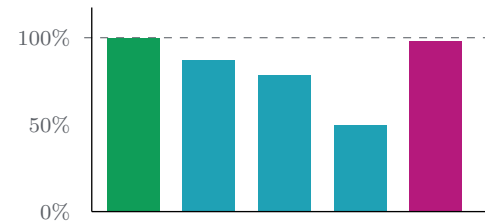


Slider = logit receptors — one scalar α steers the model.



Before & after

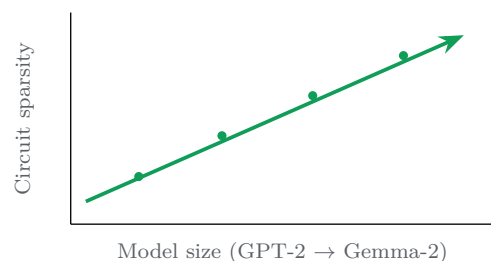
Pronoun flipped by adjusting *just one scalar*.



Ultra-compression

Prune > 90% of singular directions — task performance holds up.

4. The scaling hypothesis



Claim. Algorithmic circuits remain **localised in low-rank singular directions as models scale**. Bigger models are not more diffuse — they are *more* sparse in the SVD basis.

Why it matters for the Compass project. If this holds, the same scan-and-ablate recipe we demonstrated on GPT-2 / Phi-3 / Gemma-2 / Llama-3.2 should keep working on larger models, with equal or smaller α budgets.